INTERNATIONAL MONETARY FUND

REPUBLIC OF MOLDOVA

STAFF REPORT FOR THE 2014 ARTICLE IV CONSULTATION AND FIRST POST-PROGRAM MONITORING DISCUSSIONS—DEBT SUSTAINABILITY ANALYSIS

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Moldova's risk of debt distress remains low based on an assessment of its public external debt, but with a heightened overall risk of debt distress, reflecting vulnerabilities related to domestic debt and private external debt.[1] This finding is in line with the previous DSA, which was conducted as part of the 2012 Article IV consultation.[2] In view of the country's vulnerability to growth shocks and potential contingent liability arising from recapitalization needs of the banking system, continued fiscal discipline is critical to ensuring sustainability. There are also risks related to the high external debt to GDP ratio, emanating largely from unusually high private debt for a low-income country.

[1] This full DSA was prepared jointly by IMF and World Bank staff, in consultation with the Moldovan authorities, using the debt sustainability framework for low-income countries approved by the Boards of both institutions, and in accordance with the new staff guidance note on the application of the joint Bank-Fund Debt Sustainability Framework for Low-Income Countries (SM/13/292). The inclusion of the overall risk is in line with the new staff guidance note on the application of the joint Bank-Fund Debt Sustainability Framework for Low-Income Countries (SM/13/292). This assessment reflects the high level of external private sector debt.

[2] IMF Country Report for Moldova (No. 12/288).

BACKGROUND

1. **Moldova's total external debt at end–2013 was 84.4 percent of GDP compared to 82.4 percent of GDP at end–2012.** [3] Public and public-guaranteed (PPG) external debt decreased by about 1 percentage point and private external debt increased by 3 percentage points.[4] Moldova's PPG debt at end–2013 was about 30 percent of GDP compared to 31 percent at end–2011. PPG debt is mainly external (¾ of total) and held by multilateral and bilateral donors that hold 98 percent of PPG external debt. Domestic debt (¼ of total) is mostly short term and about 66 percent is held by the domestic banking system. In contrast, PPG external debt is mostly medium- and long-term, and the high level of concessionality of official borrowing helps keep the external debt service manageable.

2. **Private sector debt is high for a low-income country**. The stock of external private sector debt has increased in recent years, reaching US$4.9 billion as at end–2013, mostly due to increased trade credit and commercial loans. There has been rising external exposure of the banking system, mainly on a long-term basis, and bank's share of total private external debt reached about 8.7 percent by end 2013. Medium and long-term's share of total external private debt was about 55 percent as at end-2013. The majority of non-bank debt is short term, and consists of trade credits, arrears and other payments liabilities, mostly for the imports of natural resources.[5] Private borrowing in Moldova, similar to other CEE countries are extended mainly to foreign-owned companies from their parents abroad.

UNDERLYING ASSUMPTIONS

3. **The macroeconomic outlook has been revised to reflect changes in near- and medium-term projections compared to the previous DSA assessment**. In 2013, the Moldovan economy expanded at an above-trend pace with real GDP growth reaching 8.9 percent, on the back of the strong recovery in agricultural production after the previous year's severe drought. This year, however, GDP is expected to grow at 2.2 percent, mainly reflecting the anticipated slowdown in agriculture and weaker external environment. Inflation is projected to remain within the central

[3] The current DSA includes intra-company lending as private external debt. Previously, it was classified as foreign direct investment (FDI) on the presumption that it constituted a form of equity investment. As a result of this reclassification, the external-debt-to GDP ratio at end-2013 was 84.4 percent of GDP compared to near 71.9 percent projected in the 2013 Article IV report.

[4] PPG debt covers gross debt of the general government, while debts of state-owned enterprises are not included unless they are explicitly guaranteed by the government. As of September 2013, the government-guaranteed portion of SOE debt amounted to 0.1 percent of GDP, while the total amount of SOE debt, including external borrowing and domestic loans from commercial banks, stood at about 2.4 percent of GDP. In line with the DSA guidelines, public debt includes liabilities to the IMF. Small differences relative to the macro framework reflect primarily that DSA debt does not include arrears. Similarly, small differences in the primary surplus are because in the DSA it is calculated as the overall balance net of interest payments while in the macro framework tables it is calculated as the overall balance net of interest payments and earnings.

[5] The arrears are a long standing issue related to energy imports from Russia.

bank's target range as in the previous assessment. The external positions were stronger in 2012 and 2013 compared to the previous assessment, resulting in stronger external buffers. In this regard, the medium-term current account deficit is projected to be narrower than in the 2012 Article IV consultation. The fiscal position is not projected to improve as much as envisaged in the previous DSA. Over the medium–to long–term, the baseline scenario now assumes a lower potential and actual output growth (Box 1), reflecting departure from policies agreed under the previous Fund-supported program on fiscal and financial sector policies.

Evolution of Selected Macroeconomic Indicators, 2012-2017
(Percent of GDP, unless otherwise indicated)

	2012	2013	2014	2015	2016	2017
Real GDP growth (Percent)						
Previous DSA 1/	3.0	5.0	5.0	5.5	5.3	5.3
Current DSA	-0.7	8.9	2.2	3.5	4.0	4.0
Nominal GDP (Billions of U.S. dollars)						
Previous DSA 1/	7.6	8.2	9.0	9.8	10.6	11.7
Current DSA	7.3	7.9	7.6	8.1	8.7	9.2
Overall fiscal balance						
Previous DSA 1/	-1.3	-1.1	-1.0	-0.9	-0.8	-0.8
Current DSA	-2.2	-1.8	-2.6	-4.6	-4.8	-5.1
Current account balance						
Previous DSA 1/	-12.2	-11.7	-10.9	-10.4	-9.9	-9.5
Current DSA	-7.7	-5.5	-7.4	-8.2	-8.1	-8.0

Sources: IMF Staff Estimates and calculations.

1/ See IMF Country Report No. 12/288.

EXTERNAL AND PUBLIC DEBT SUSTAINABILITY

A. External Debt Sustainability Analysis

4. **All external public debt ratios remain well below the thresholds under the baseline and stress scenarios**. In the alternative scenario in which key variables remain at their historical averages (2013–2033), the present value of debt-to-GDP and remittances ratio would reach 21 percent compared to 13 percent under the baseline scenario. The present value of debt-to-exports and remittances as well as to revenue would also increase substantially. This result is an indication of the significant fiscal and external adjustments that have taken place in recent years. At the same time, it underscores the importance for Moldova of staying on the path of sound economic policies and reforms.

Box 1. Macroeconomic Assumptions behind the DSA

Real GDP increased by 8.9 percent in 2013 and is projected to grow at 2.2 percent in 2014 driven by the anticipated slowdown in agriculture. An expected recovery in FDI following the signing of the DCFTA with the EU as well as advances in structural reforms will help sustain the economy's potential growth rate at 4 percent in the medium term.

Inflation is expected to average 5½–6 percent in 2014–15, reflecting robust domestic demand combined with increased utility tariffs. Headline inflation is expected to be 5 percent over the medium to long term in line with the NBM' mid-target.

Exports are expected to be a key source of economic growth. They are projected to further increase in the medium term in line with the recent trends and supported by structural reforms to improve the business environment, reduce the costs of doing business and enhance the provision of infrastructure. As a result, exports of goods and services are projected to reach 49.8 percent of GDP in 2033 (compared with 44.1 percent in 2013).

Imports are projected to grow as well, underpinned by intermediate imports needed for the exports, though the decline in the growth of remittances would contribute to moderating the growth of imports. Imports as a ratio of GDP are projected at 84percent of GDP in 2033.

Remittances grew rapidly in 2012 and 2013 due to construction-related activities in Russia and the social challenges associated with drought in Moldova in 2012. The growth of remittances is envisaged to slow in the medium term. In the long term, also, as the economy develops, more employment options are available domestically, and migrants abroad lose ties with the home country, remittances are projected to decline relative to GDP.

The current account deficit is projected to narrow slightly in the medium term, following the projected deterioration in 2014 and 2015, emanating largely from marked slowdown in the growth of remittances and stabilize at about 7.7 percent of GDP. It will be financed by FDI that is expected to reach close to about 4½ percent of GDP in the long run.

The overall budget balance is projected to be a deficit of 2.6 percent of GDP in 2014 and to increase to 4.7 percent by 2019, under the baseline scenario. Over the long run (2020–2033), the primary budget balance is assumed to be an average deficit of 2.9 percent of GDP.

Financing assumptions reflect a gradual shift away from concessional financing. Grant equivalent financing is projected to decline from 2.1 percent of GDP in 2018 to 1.4 percent in 2033. At the same time, the grant element of new borrowing is projected to fall from 37 percent to 22 percent during the same period.

5. **While fiscal consolidation, higher exports, and significant levels of remittances should ensure adequate resources for external public debt service, risks still exist**.[6] While none of the indicators of debt service breaches the threshold, some liquidity pressures could emerge. Significant private external debt and its short maturity represent a risk in case roll-over-rates were to decline sharply. On the positive side, the improvement in the current account position in 2012 and 2013 has allowed the NBM to build-up reserves, enhancing the economy's resilience to adverse exogenous shocks.

B. Public Sector Debt Sustainability Analysis

6. **Public debt dynamics appear to be sustainable under the baseline scenario, but the level of PPG total debt is projected to increase over the medium term**. With the current set of macroeconomic policies outlined above, the primary budget deficit path is projected to be wider than the level that would stabilize the debt-to-GDP ratio. Accordingly, Moldova's PPG total debt is expected to increase from 29.8 percent of GDP at the end of 2013 to 39.4 percent by end–2018 and 54.5 percent by 2033. Other sustainability indicators confirm this negative long-term trend under the baseline scenario. The PV of debt-to-GDP is projected to increase from 24.2 percent in 2013 to 32.6 percent in 2018 and 49 percent in 2033, albeit below the benchmark level of 56 percent.[7] Similarly, the PV of debt-to-revenue and grants ratio is projected to deteriorate from 65.4 percent in 2013 to 89.4 percent in 2018 and 134.6 percent by 2033.

7. **Moldova's public debt trajectory appears to be particularly vulnerable to growth and primary balance shocks**. Although Moldova's risk of debt distress remains low under the baseline scenario, standard alternative scenarios presented in the DSA framework highlight a number of significant vulnerabilities. For example, a permanent one standard deviation in real GDP growth under the baseline scenario would increase the PV of debt-to-GDP to 58 percent by 2033 and 107 percent in 2033, breaching the 56 percent benchmark level. Therefore, while the DSA framework points to a low risk of debt distress over the period 2014–19, stress tests indicate that the debt sustainability is vulnerable to exogenous (growth) shocks and potential fiscal risks. These results underscore the importance of maintaining prudent fiscal policies over the medium term.

8. **Potential recapitalization needs of the banking system represent an additional risk for the PPG total debt**. If the cost of bank recapitalization were 10 percent of GDP, the stock of public debt would increase to 41.8 percent of GDP in 2014. Although the PV of debt-to-GDP would be 32.5 percent in 2014, it would increase to 52 percent by 2033, marginally below the benchmark level of 56 percent (see Table 2a, Bound Test 5).

[6] In Moldova, remittances are classified as either current transfers or compensation of employees. Both categories are included in the DSA (under "current transfers").

[7] See Revisiting the Debt Sustainability Framework for Low-Income Countries for a discussion of public debt benchmarks. Moldova's three-year average score on the Bank's Country Policy and Institutional Assessment (CPIA) is 3.77, which places the country at the upper-end of the medium policy performance category.

C. The Authorities' View

9. **The authorities concurred with the staff assessment, and agreed that the introduction of a rule-based fiscal policy framework would help alleviate risks.** The authorities acknowledged that the current fiscal position (as discussed with the IMF 2014 Article IV mission in April 2014) does not improve debt sustainability. As a response, they proposed to introduce fiscal rules to anchor fiscal policy and thereby ensure debt sustainability over the long term. The authorities also noted that short-term debt is mostly related to trade credits and indicated that the country's ambitious development plan requires a sustained increase in borrowing. In this context, they would like to develop a long-term debt management strategy with a focus on exploring alternative sources of financing while maintaining a sustainable debt level. As part of the strategy to develop the domestic debt market, the authorities pointed out that the Treasury has started issuing long-term government bonds and that the NCFM has approved the trading of government securities (with maturity over one year) on the stock exchange.

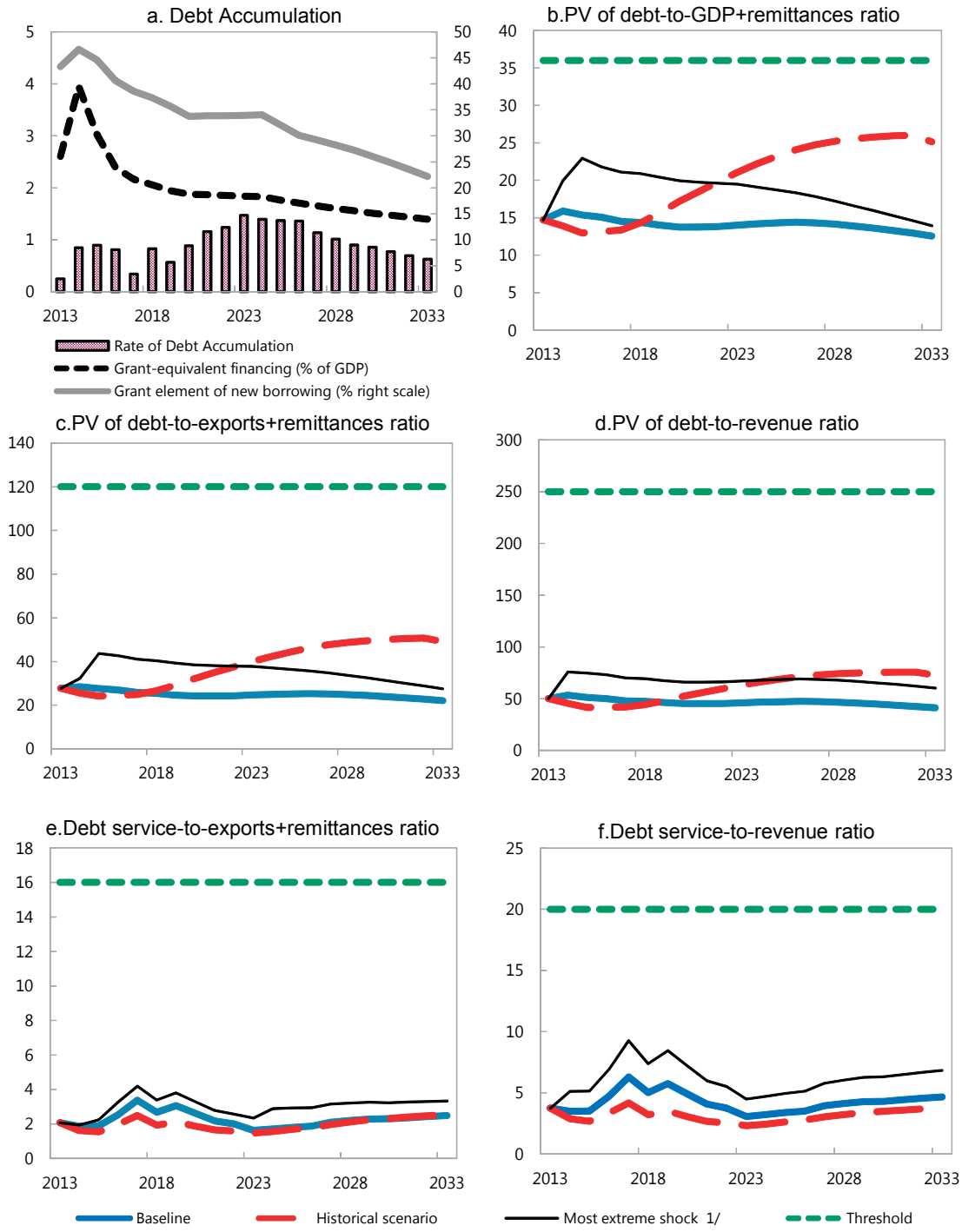
CONCLUSION

10. **The DSA indicates that Moldova's risk of debt distress remains low, in line with the 2012 assessment, but with heightened overall risk**. All external indicators for public debt remain well under the thresholds under the standard bound tests and alternative scenarios. However, significant private external debt poses roll over risks to debt sustainability. Similarly, although Moldova's public debt dynamics are projected to remain on a sustainable path under the baseline scenario, standard alternative scenarios presented in this assessment highlight particular sensitivity to assumptions on output growth and the path of fiscal policy.

11. **Pursuing prudent fiscal policy and advancing structural reforms are necessary to ensure debt sustainability**. In view of Moldova's sensitivity to exogenous developments, the debt sustainability critically depends on sound macroeconomic management and continued progress on institutional and structural issues that would help unlock the economy's growth potential and reduce its vulnerability to shocks. Furthermore, the limited development of the domestic debt market poses financing risks, especially considering the country's developments needs and significant dependence on foreign assistance in the form of grants and concessional loans. In this context, efforts to lengthen average maturity of domestic debt and deepen the secondary market would help reduce the PPG domestic debt rollover and interest rate risks.

12. **A reduction or reversal of trade credit and other private financial inflows could have an adverse impact on the economy**. A possible decline in banking sector liquidity in foreign countries or deterioration in banking sector confidence could lead to reduced availability of trade financing. In addition, deterioration in the economic conditions in Moldova's main economic partners—where the parent companies are located—could lead to some decline in external financing for the private sector. All these points to roll-over risks for the private sector.

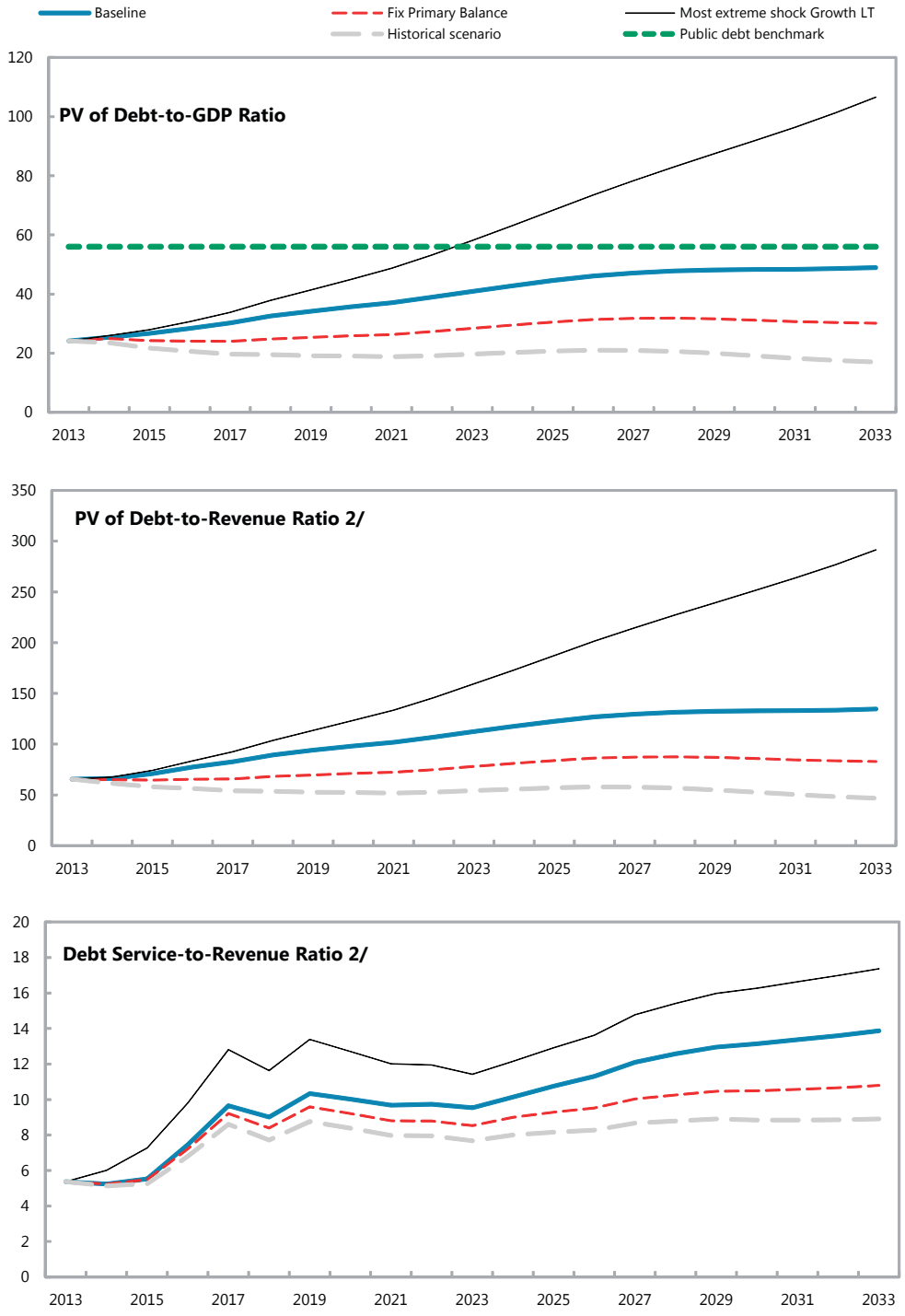
Figure 1. Moldova: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2013–2033 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2023. In figure b. it corresponds to a Non-debt flows shock; in c. to a Exports shock; in d. to a One-time depreciation shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Moldova: Indicators of Public Debt Under Alternative Scenarios, 2013–2033 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2023.
2/ Revenues are defined inclusive of grants.

Table 1. Moldova: Public Sector Debt Sustainability Framework, Baseline Scenario, 2010–2033

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate		Projections							
	2010	2011	2012	Average 5/	Standard Deviation 5/	2013	2014	2015	2016	2017	2018	2013-18 Average	2023	2033	2019-33 Average
Public sector debt 1/	30.2	28.8	31.0			29.8	31.8	32.8	34.7	36.7	39.4		48.5	54.8	
of which: foreign-currency denominated	22.8	21.7	24.0			23.1	25.3	24.4	24.0	23.6	23.6		23.9	20.4	
Change in public sector debt	-1.5	-1.4	2.2			-1.2	2.0	1.0	1.9	2.1	2.6		2.0	0.1	
Identified debt-creating flows	-3.3	-2.6	0.5			-0.4	2.6	1.7	2.2	2.5	2.2		1.0	1.1	
Primary deficit	1.7	1.6	1.4	-0.1	2.5	1.2	1.9	3.9	3.9	3.9	3.5	3.1	2.9	3.0	2.9
Revenue and grants	38.3	36.6	37.9			37.0	38.4	37.6	36.7	36.5	36.4		36.4	36.4	
of which: grants	2.8	2.1	1.8			2.1	2.9	2.0	1.3	1.1	1.0		1.0	1.0	
Primary (noninterest) expenditure	40.0	38.2	39.3			38.1	40.3	41.4	40.6	40.4	39.9		39.2	39.3	
Automatic debt dynamics	-4.5	-3.7	-0.6			-1.3	0.9	-2.0	-1.5	-1.2	-1.2		-1.8	-1.9	
Contribution from interest rate/growth differential	-2.3	-2.0	0.1			-2.5	-0.7	-1.1	-1.2	-1.1	-1.2		-1.7	-1.8	
of which: contribution from average real interest rate	-0.2	0.0	-0.1			0.0	-0.1	0.0	0.0	0.2	0.2		0.5	0.8	
of which: contribution from real GDP growth	-2.1	-1.9	0.2			-2.5	-0.6	-1.1	-1.3	-1.3	-1.4		-2.2	-2.6	
Contribution from real exchange rate depreciation	-2.2	-1.7	-0.7			1.2	1.7	-0.9	-0.2	-0.1	0.0		...	...	
Other identified debt-creating flows	-0.5	-0.4	-0.3			-0.3	-0.3	-0.2	-0.2	-0.2	-0.2		-0.1	0.0	
Privatization receipts (negative)	-0.5	-0.4	-0.3			-0.3	-0.3	-0.2	-0.2	-0.2	-0.2		-0.1	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	1.8	1.2	1.7			-0.8	-0.6	-0.6	-0.4	-0.4	0.4		1.0	-1.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	25.0			24.2	25.5	26.7	28.4	30.2	32.6		40.9	49.0	
of which: foreign-currency denominated	...	...	18.0			17.5	19.0	18.2	17.8	17.0	16.8		16.3	14.6	
of which: external	...	...	18.0			17.5	19.0	18.2	17.8	17.0	16.8		16.3	14.6	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	11.0	10.2	10.0			9.2	9.9	11.7	14.1	16.8	18.5		26.3	37.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	65.9			65.4	66.2	71.0	77.4	82.7	89.4		112.5	134.6	
PV of public sector debt-to-revenue ratio (in percent)	...	...	69.2			69.3	71.6	75.0	80.2	85.3	92.0		115.6	138.3	
of which: external 3/	...	...	49.9			50.1	53.4	51.3	50.2	48.1	47.6		46.2	41.3	
Debt service-to-revenue and grants ratio (in percent) 4/	6.0	6.4	5.7			5.4	5.2	5.5	7.5	9.7	9.0		9.5	13.9	
Debt service-to-revenue ratio (in percent) 4/	6.5	6.8	6.0			5.7	5.7	5.9	7.7	10.0	9.3		9.8	14.3	
Primary deficit that stabilizes the debt-to-GDP ratio	3.2	2.9	-0.8			2.4	-0.1	2.8	2.1	1.8	0.9		0.8	2.9	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.1	6.8	-0.7	4.4	4.5	8.9	2.2	3.5	4.0	4.0	4.0	4.4	5.0	5.0	4.9
Average nominal interest rate on forex debt (in percent)	1.3	1.2	1.0	1.7	0.4	1.1	1.0	1.1	1.2	1.8	1.8	1.3	2.0	2.2	2.1
Average real interest rate on domestic debt (in percent)	-3.4	1.8	0.6	0.6	6.7	1.2	0.3	2.0	2.3	2.5	2.4	1.8	2.3	2.4	2.4
Real exchange rate depreciation (in percent, + indicates depreciation)	-10.1	-8.2	-3.1	-7.8	10.4	5.6	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	11.1	7.3	7.9	9.9	4.1	4.0	6.5	5.6	5.7	5.5	5.1	5.4	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.0	0.0	0.1	0.1	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.0
Grant element of new external borrowing (in percent)	...	...	...	...	...	43.3	46.7	44.6	40.6	38.5	37.3	41.8	33.9	22.2	...

Sources: Country authorities; and staff estimates and projections.

1/ Gross public and publicly-guaranteed debt, excluding arrears. The primary budget balance in this analysis is defined as the overall balance net of interest payments.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2. Moldova: Sensitivity Analysis for Key Indicators of Public Debt 2013–2033

	Projections							
	2013	2014	2015	2016	2017	2018	2023	2033
PV of Debt-to-GDP Ratio								
Baseline	24	25	27	28	30	33	41	49
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	24	24	22	21	20	20	20	17
A2. Primary balance is unchanged from 2013	24	25	24	24	24	25	28	30
A3. Permanently lower GDP growth 1/	24	26	28	31	34	38	58	107
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	24	27	30	34	37	41	56	74
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	24	26	26	28	30	32	40	49
B3. Combination of B1-B2 using one half standard deviation shocks	24	25	25	27	29	32	42	53
B4. One-time 30 percent real depreciation in 2014	24	35	35	36	37	39	45	53
B5. 10 percent of GDP increase in other debt-creating flows in 2014	24	33	33	35	37	39	47	52
PV of Debt-to-Revenue Ratio 2/								
Baseline	65	66	71	77	83	89	113	135
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	65	61	58	56	54	54	54	47
A2. Primary balance is unchanged from 2013	65	65	65	66	66	68	78	83
A3. Permanently lower GDP growth 1/	65	67	74	84	93	104	159	291
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	65	69	81	92	101	112	154	203
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	65	67	69	75	81	88	111	134
B3. Combination of B1-B2 using one half standard deviation shocks	65	65	65	73	79	87	115	146
B4. One-time 30 percent real depreciation in 2014	65	90	93	97	101	106	124	147
B5. 10 percent of GDP increase in other debt-creating flows in 2014	65	85	89	95	100	107	128	144
Debt Service-to-Revenue Ratio 2/								
Baseline	5	5	6	7	10	9	10	14
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	5	5	5	7	9	8	8	9
A2. Primary balance is unchanged from 2013	5	5	5	7	9	8	9	11
A3. Permanently lower GDP growth 1/	5	5	6	8	10	10	11	22
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2014-2015	5	5	6	8	11	10	11	18
B2. Primary balance is at historical average minus one standard deviations in 2014-2015	5	5	6	7	10	9	9	14
B3. Combination of B1-B2 using one half standard deviation shocks	5	5	6	7	10	9	10	14
B4. One-time 30 percent real depreciation in 2014	5	6	7	10	13	12	11	17
B5. 10 percent of GDP increase in other debt-creating flows in 2014	5	5	6	8	10	10	10	15

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table 3a. Moldova: External Debt Sustainability Framework, Baseline Scenario, 2010–2033 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2010	2011	2012			2013	2014	2015	2016	2017	2018	2013-2018 Average	2023	2033	2019-2033 Average
External debt (nominal) 1/	**81.6**	**77.6**	**82.4**			**84.4**	**90.8**	**87.9**	**86.3**	**85.0**	**84.4**		**78.5**	**68.0**	
of which: public and publicly guaranteed (PPG)	22.8	21.7	24.0			23.1	25.3	24.4	24.0	23.6	23.6		23.9	20.4	
Change in external debt	-0.8	-4.1	4.8			2.0	6.4	-2.9	-1.6	-1.3	-0.6		-1.2	-1.0	
Identified net debt-creating flows	0.7	-5.8	2.5			-3.9	2.7	1.6	0.7	0.3	-0.1		-0.6	-0.2	
Non-interest current account deficit	**8.7**	**11.3**	**6.8**	**8.8**	**4.8**	**4.8**	**5.6**	**6.1**	**6.0**	**5.8**	**5.6**		**5.4**	**6.0**	5.4
Deficit in balance of goods and services	39.3	40.9	40.4			37.6	40.8	38.9	36.8	35.8	34.9		34.3	34.3	
Exports	39.4	45.0	43.4			44.1	47.2	47.3	47.9	48.4	49.2		49.8	49.8	
Imports	78.7	85.9	83.9			81.7	88.1	86.2	84.7	84.2	84.1		84.0	84.0	
Net current transfers (negative = inflow)	-21.0	-20.5	-21.2	-20.9	4.1	-20.9	-22.4	-20.8	-19.5	-19.0	-18.6		-18.1	-18.1	-18.1
of which: official	-2.3	-2.0	-2.2			-2.1	-2.8	-2.3	-1.6	-1.6	-1.6		-1.6	-1.6	
Other current account flows (negative = net inflow)	-9.6	-9.2	-12.4			-11.9	-12.8	-12.0	-11.3	-11.0	-10.7		-10.8	-10.2	
Net FDI (negative = inflow)	**-3.6**	**-4.1**	**-2.3**	**-5.9**	**3.6**	**-2.8**	**-2.7**	**-3.6**	**-4.0**	**-4.4**	**-4.8**		**-4.7**	**-4.8**	-4.7
Endogenous debt dynamics 2/	**-4.4**	**-13.0**	**-1.9**			**-5.9**	**-0.2**	**-0.9**	**-1.2**	**-1.0**	**-0.9**		**-1.3**	**-1.4**	
Contribution from nominal interest rate	0.9	1.0	0.9			0.8	1.8	2.1	2.1	2.2	2.3		2.4	1.8	
Contribution from real GDP growth	-5.5	-4.6	0.5			-6.7	-1.9	-3.0	-3.3	-3.2	-3.2		-3.7	-3.2	
Contribution from price and exchange rate changes	0.1	-9.4	-3.4			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-1.5**	**1.8**	**2.3**			**6.0**	**3.7**	**-4.6**	**-2.3**	**-1.6**	**-0.5**		**-0.6**	**-0.8**	
of which: exceptional financing	-2.4	-1.9	-3.2			-1.9	-2.4	-1.4	-0.9	-0.7	-0.8		-0.4	-0.2	
PV of external debt 4/	...	...	76.4			78.8	84.5	81.7	80.0	78.5	77.6		70.9	62.2	
In percent of exports	...	...	175.8			178.6	179.0	172.8	167.2	162.1	157.6		142.6	125.0	
PV of PPG external debt	**...**	**...**	**18.0**			**17.5**	**19.0**	**18.2**	**17.8**	**17.0**	**16.8**		**16.3**	**14.6**	
In percent of exports	**...**	**...**	**41.5**			**39.6**	**40.2**	**38.6**	**37.1**	**35.2**	**34.2**		**32.9**	**29.4**	
In percent of government revenues	**...**	**...**	**49.9**			**50.1**	**53.4**	**51.3**	**50.2**	**48.1**	**47.6**		**46.2**	**41.3**	
Debt service-to-exports ratio (in percent)	**17.8**	**15.9**	**15.5**			**17.4**	**18.4**	**18.6**	**21.0**	**19.7**	**19.9**		**18.0**	**17.6**	
PPG debt service-to-exports ratio (in percent)	**3.4**	**3.0**	**2.9**			**3.0**	**2.6**	**2.6**	**3.5**	**4.6**	**3.6**		**2.2**	**3.3**	
PPG debt service-to-revenue ratio (in percent)	**3.8**	**3.9**	**3.5**			**3.7**	**3.5**	**3.5**	**4.7**	**6.3**	**5.0**		**3.1**	**4.7**	
Total gross financing need (Millions of U.S. dollars)	2037	2471	2592			2596	3090	3143	3307	3384	3558		4404	7645	
Non-interest current account deficit that stabilizes debt ratio	9.5	15.4	2.0			2.7	-0.8	9.0	7.6	7.0	6.2		6.6	6.9	
Key macroeconomic assumptions															
Real GDP growth (in percent)	7.1	6.8	-0.7	4.4	4.5	8.9	2.2	3.5	4.0	4.0	4.0	4.4	5.0	5.0	4.9
GDP deflator in US dollar terms (change in percent)	-0.2	13.0	4.6	11.6	10.7	0.1	-5.9	2.7	3.0	2.3	2.0	0.7	2.3	2.3	2.3
Effective interest rate (percent) 5/	1.2	1.5	1.2	1.6	0.3	1.0	2.0	2.5	2.5	2.7	2.9	2.3	3.2	2.8	3.3
Growth of exports of G&S (US dollar terms, in percent)	14.5	37.9	0.2	14.3	18.0	10.6	2.9	6.5	8.4	7.6	7.8	7.3	7.4	7.4	7.4
Growth of imports of G&S (US dollar terms, in percent)	14.7	31.9	1.3	17.1	21.0	6.1	3.7	4.1	5.2	5.9	5.9	5.1	7.4	7.4	7.3
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	43.3	46.7	44.6	40.6	38.5	37.3	41.8	33.9	22.2	29.9
Government revenues (excluding grants, in percent of GDP)	35.5	34.5	36.1			34.9	35.6	35.5	35.4	35.4	35.4		35.4	35.4	35.4
Aid flows (in Millions of US dollars) 7/	225.6	200.3	206.5			196.5	299.5	233.1	180.3	174.0	177.7		214.4	353.1	
of which: Grants	161.7	145.2	130.6			164.7	218.4	165.6	113.8	100.3	99.6		133.3	272.0	
of which: Concessional loans	63.9	55.1	75.9			31.8	81.1	67.5	66.5	73.7	78.1		81.1	81.1	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			2.6	4.0	3.0	2.4	2.2	2.1		1.8	1.4	1.7
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			79.1	76.1	71.3	60.2	55.7	54.0		51.8	47.7	50.2
Memorandum items:															
Nominal GDP (Millions of US dollars)	5813	7018	7283			7935	7628	8109	8686	9242	9803		13872	28315	
Nominal dollar GDP growth	6.9	20.7	3.8			8.9	-3.9	6.3	7.1	6.4	6.1	5.2	7.4	7.4	7.3
PV of PPG external debt (in Millions of US dollars)			1319			1336	1403	1471	1537	1566	1642		2257	4121	
(PVt-PVt-1)/GDPt-1 (in percent)						0.2	0.8	0.9	0.8	0.3	0.8	0.7	1.5	0.6	1.0
Gross workers' remittances (Millions of US dollars)	1083	1299	1389			1495	1493	1503	1557	1612	1666		2285	4664	
PV of PPG external debt (in percent of GDP + remittances)	...	...	15.1			14.7	15.9	15.4	15.1	14.5	14.4		14.0	12.6	
PV of PPG external debt (in percent of exports + remittances)	...	...	28.8			27.8	28.5	27.7	27.0	25.9	25.4		24.7	22.1	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.0			2.1	1.9	1.9	2.5	3.4	2.7		1.6	2.5	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt. The difference between end-period and average exchange rates account for the difference in external debt ratio presented in Table 1 (SEI) and the DSA.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 3b. Moldova: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–2033

(In percent)

	Projections							
	2013	2014	2015	2016	2017	2018	**2023**	2033
PV of debt-to-GDP+remittances ratio								
Baseline	15	16	15	15	15	14	**14**	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	15	14	13	12	13	13	**18**	21
A2. New public sector loans on less favorable terms in 2013-2033 2	15	16	16	16	16	17	**19**	22
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	15	16	16	16	15	15	**15**	13
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	15	17	22	21	20	20	**19**	14
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	15	15	15	14	14	14	**13**	12
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	15	20	23	22	21	21	**19**	14
B5. Combination of B1-B4 using one-half standard deviation shocks	15	13	11	11	11	11	**11**	11
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	15	21	21	21	20	20	**19**	17
PV of debt-to-exports+remittances ratio								
Baseline	28	28	28	27	26	25	**25**	22
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	28	25	23	23	24	25	**35**	41
A2. New public sector loans on less favorable terms in 2013-2033 2	28	29	29	30	29	30	**34**	38
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	28	28	28	27	26	25	**25**	22
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	28	32	44	43	41	40	**38**	28
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	28	28	28	27	26	25	**25**	22
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	28	37	42	39	38	37	**34**	25
B5. Combination of B1-B4 using one-half standard deviation shocks	28	26	23	23	22	21	**21**	21
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	28	28	28	27	26	25	**25**	22
PV of debt-to-revenue ratio								
Baseline	50	53	51	50	48	48	**46**	41
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	50	45	40	40	39	42	**55**	61
A2. New public sector loans on less favorable terms in 2013-2033 2	50	54	54	55	55	56	**64**	71
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	50	53	54	53	51	50	**49**	43
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	50	58	72	70	68	67	**63**	46
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	50	48	48	47	45	45	**44**	39
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	50	64	74	73	70	69	**64**	46
B5. Combination of B1-B4 using one-half standard deviation shocks	50	42	36	36	34	34	**34**	34
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	50	76	75	73	70	69	**67**	60

Table 3b. Moldova: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2013–2033 (concluded)

(In percent)

Debt service-to-exports+remittances ratio								
Baseline	2	2	2	3	3	3	**2**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	2	2	2	2	2	2	**1**	2
A2. New public sector loans on less favorable terms in 2013-2033 2	2	2	2	3	4	3	**2**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	2	2	2	3	3	3	**2**	2
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	2	2	2	3	4	3	**2**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	2	2	2	3	3	3	**2**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	2	2	2	3	4	3	**2**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	2	2	3	3	**1**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	2	2	2	3	3	3	**2**	2
Debt service-to-revenue ratio								
Baseline	4	3	3	5	6	5	**3**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2013-2033 1/	4	3	3	3	4	3	**2**	3
A2. New public sector loans on less favorable terms in 2013-2033 2	4	3	4	5	7	5	**3**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2014-2015	4	4	4	5	7	5	**3**	5
B2. Export value growth at historical average minus one standard deviation in 2014-2015 3/	4	3	4	5	7	6	**4**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2014-2015	4	3	3	4	6	5	**3**	4
B4. Net non-debt creating flows at historical average minus one standard deviation in 2014-2015 4/	4	3	4	5	7	6	**4**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	4	3	3	4	5	4	**2**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2014 5/	4	5	5	7	9	7	**4**	7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	27	27	27	27	27	27	**27**	27

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.